Numerical simulation of the Izhikevich neuron model

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Abstract. The mathematical modelling of neurons requires balancing the realism of the description with the simplicity needed for analysis and simulation. The Izhikevich model lies at that middle point, since with two differential equations and four parameters it reproduces a wide variety of firing patterns at a low computational cost. In this work, this model was implemented and analysed, integrated with the fourth-order Runge–Kutta method (RK4) with step $h = 0.1$ ms and a reset condition applied when $v \geq 30$ mV. First, the response of a neuron with typical parameters to a current step was studied. Then, eight patterns of neuronal activity were reproduced (RS, IB, CH, FS, LTS, TC1, TC2 and RZ). The simulations qualitatively reproduce the patterns of the original paper and show that their diversity depends on both the model parameters and the input current. Finally, the effect of the integration step was analysed. Spike times depend on h and their error accumulates with each spike because of the reset, although the qualitative patterns are not altered.

Keywords: Izhikevich model, numerical simulation, differential equations, neuronal dynamics

I. INTRODUCTION

Mathematical and computational modelling is a fundamental tool for studying how the nervous system works, since it makes it possible to describe neuronal activity and to explore the mechanisms that underlie it. As Dayan and Abbott [1] point out, a more detailed model is not necessarily better; rather, it should act as a bridge between different levels of understanding, incorporating enough information to represent the biological mechanisms of interest while keeping the simplicity needed to facilitate its analysis.

This balance between detail and simplicity is reflected in spiking neuron models, which describe the evolution of the membrane potential and the generation of action potentials in response to input currents. Integrate-and-Fire models, derived from Lapicque's proposal [2], represent the integration of the input currents and the firing by means of a threshold, followed by a reset of the membrane potential, without describing the action potential in detail. This facilitates mathematical analysis and reduces the computational cost of simulations [1]. The Hodgkin–Huxley model, on the other hand, incorporates voltage-dependent sodium and potassium conductances, which makes it possible to describe the generation and time course of the action potential, although it requires solving a more complex system of equations [3].

In this context, the Izhikevich model combines the richness of neuronal dynamics that characterises the Hodgkin–Huxley model with the computational efficiency of Integrate-and-Fire models. Its formulation is based on bifurcation theory and on the reduction to normal forms, which makes it possible to preserve relevant features of neuronal dynamics through a simplified description. With two differential equations and a reset rule, it reproduces different firing and bursting patterns by tuning four parameters, which facilitates the simulation of large neural networks [4].

The aim of this work is to study the ability of the Izhikevich model to reproduce different patterns of neuronal activity through a simplified mathematical description. To this end, the model was simulated with the fourth-order Runge-Kutta method, the way in which the parameters and the input currents determine neuronal behaviour was analysed, the patterns obtained were compared with those of the original paper, and the effect of the integration step on spike times was evaluated.

II. THEORY AND NUMERICAL METHODS

A. The Izhikevich model

The model is described by the following system of two ordinary differential equations (ODEs):

$$\frac{dv}{dt} = 0.04\,v^2 + 5v + 140 - u + I, \tag{1}$$

$$\frac{du}{dt} = a\,(bv - u), \tag{2}$$

together with an after-spike reset mechanism:

$$\text{if } v \geq 30 \text{ mV, then } \begin{cases} v \leftarrow c, \\ u \leftarrow u + d. \end{cases} \tag{3}$$

In these equations, time t is measured in milliseconds (ms) and the membrane potential v in millivolts (mV).

B. Interpretation of the variables and the parameters

The dynamics of the model is described by two state variables, an input current and four parameters that regulate neuronal behaviour [4].

- v *(Membrane potential):* corresponds to the electrical potential difference across the membrane of the neuron and is expressed in millivolts (mV).

- *u (Recovery variable):* summarises the activation of potassium (K^+) currents and the inactivation of sodium (Na^+) currents. It provides negative feedback on the membrane potential.

- *I (Input current):* represents the current applied to the neuron, which may be injected directly or correspond to synaptic currents. In this work it is a function of time, and its versions I_0–I_4 are defined in Table I.

- *a (Time scale of u):* determines how fast the recovery variable u evolves. Smaller values produce slower recovery. A typical value is $a = 0.02$.

- *b (Sensitivity of u to v):* describes how the variable u responds to the subthreshold fluctuations of v. Larger values couple both variables more strongly and may favour subthreshold oscillations and low-threshold spiking. A typical value is $b = 0.2$.

- *c (Potential reset):* sets the value to which v is reset after reaching the spike peak (+30 mV). It is related to the action of fast-activating potassium conductances. A typical value is $c = -65$ mV.

- *d (Recovery jump):* determines the instantaneous increment added to the variable u after reaching the spike peak. It is related to slower sodium and potassium conductances. A typical value is $d = 2$.

C. Cutoff condition and threshold

The value of +30 mV is not a fixed physiological firing threshold; it is used as a reference to detect the peak of the action potential and to apply the reset condition.

Unlike traditional Integrate-and-Fire (IF) models, which usually use a fixed threshold, the Izhikevich model has a dynamic effective firing threshold, which depends on the previous state of v and u and can vary approximately between −55 mV and −40 mV [4].

D. Diversity of firing patterns

Through different combinations of the parameters, the model can reproduce firing patterns characteristic of different types of cortical and thalamic neurons [4]. These include the excitatory neurons Regular Spiking (RS), Intrinsically Bursting (IB) and Chattering (CH); the inhibitory interneurons Fast Spiking (FS) and Low-Threshold Spiking (LTS); the thalamo-cortical regimes of tonic firing (TC1) and rebound bursting (TC2); and the Resonator pattern (RZ) (Figure 1).

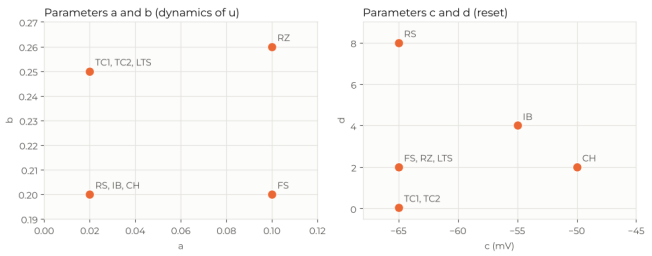


Figure 1. Location of the eight configurations used in the parameter space of the Izhikevich model.

E. Numerical integration

Equations (1) and (2) were integrated numerically with the fourth-order Runge-Kutta method (RK4), implemented in Python, with an integration step of $h = 0.1$ ms. Given the state $\mathbf{x} = (v, u)$ at time t, the state at $t + h$ is computed as:

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \frac{h}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right), \qquad (4)$$

where $\mathbf{k}_1, \ldots, \mathbf{k}_4$ are evaluations of $\mathbf{f}$, the right-hand side of Eqs. (1) and (2), at the beginning, at two intermediate points and at the end of each step.

After each integration step, it is checked whether $v \geq 30$ mV; if so, the spike time is recorded and the reset (3) is applied immediately. For the figures, the point $(t, 30\,\text{mV})$ is added at each spike before the reset value, since the integrator does not store the peak.

The initial conditions were $u(0) = b\,v(0)$, with the values of $v(0)$ given in Table I. The integration interval was $[0, 100]$ ms for the base case and $[0, 200]$ ms for the patterns, except for RZ, whose interval was $[0, 300]$ ms because its pulse occurs at 180 ms.

To evaluate the numerical error, the base case was repeated with $h = 0.2$, 0.05 and 0.01 ms, taking $h = 0.01$ ms as the reference.

III. RESULTS

A. Neuronal response to a current step

The response of the base case to a current step applied at $t = 10$ ms was analysed first. Figure 2 shows the evolution of the membrane potential and of the recovery variable, together with the input current. Until the stimulus is applied, the neuron remains at rest ($v = -70$ mV). A train of spikes is then observed, the first at $t = 13.5$ ms, whose intervals between consecutive spikes increase progressively from 3.6 ms to about 19 ms until an approximately periodic regime is reached. Each spike is accompanied by an increment of u of value $d = 2$, followed by a slow relaxation of u until the next spike. This accumulation of u is what progressively lengthens the intervals between spikes.

Table I. Parameters, initial potential and input current of the base case and of the eight simulated patterns. In all cases, $u(0) = b \cdot v(0)$.

Case	a	b	c (mV)	d	$v(0)$ (mV)	Current (t in ms)
Base case	0.02	0.20	−65	2	−70	I_0 0 ($t < 10$); 10 ($t \geq 10$)
RS	0.02	0.20	−65	8	−65	I_1 0 ($t < 20$); 10 ($t \geq 20$)
IB	0.02	0.20	−55	4	−65	I_1 0 ($t < 20$); 10 ($t \geq 20$)
CH	0.02	0.20	−50	2	−65	I_1 0 ($t < 20$); 10 ($t \geq 20$)
FS	0.10	0.20	−65	2	−65	I_1 0 ($t < 20$); 10 ($t \geq 20$)
TC1	0.02	0.25	−65	0.05	−63	I_2 0 ($t < 20$); 1 ($t \geq 20$)
TC2	0.02	0.25	−65	0.05	−65	I_3 0 ($t < 20$); −25 ($20 \leq t < 80$); 0 ($t \geq 80$)
RZ	0.10	0.26	−65	2	−62.5	I_4 0 ($t < 40$); 0.2 ($40 \leq t < 180$); 2.2 ($180 \leq t < 181$); 0.2 ($t \geq 181$)
LTS	0.02	0.25	−65	2	−65	I_1 0 ($t < 20$); 10 ($t \geq 20$)

B. Neuronal dynamics for different configurations

Figure 3 presents the response of the eight parameter configurations corresponding to the different neuron types. RS, IB, CH, FS and LTS receive the same current step ($I_1 = 10$) from 20 ms onwards, so their different responses are due exclusively to the change in the model parameters. RS, FS and LTS produce single spikes, with different frequencies and time courses. RS fires at low frequency; after a shorter first interval, the following ones become longer and keep a constant period. LTS, in contrast, starts firing at high frequency and its intervals grow progressively until they stabilise. FS has the highest frequency and keeps practically constant intervals throughout the stimulus. IB and CH, for their part, respond with bursts. In IB there is an initial burst followed by isolated spikes at regular intervals, whereas in CH the bursts repeat at approximately regular intervals throughout the stimulus.

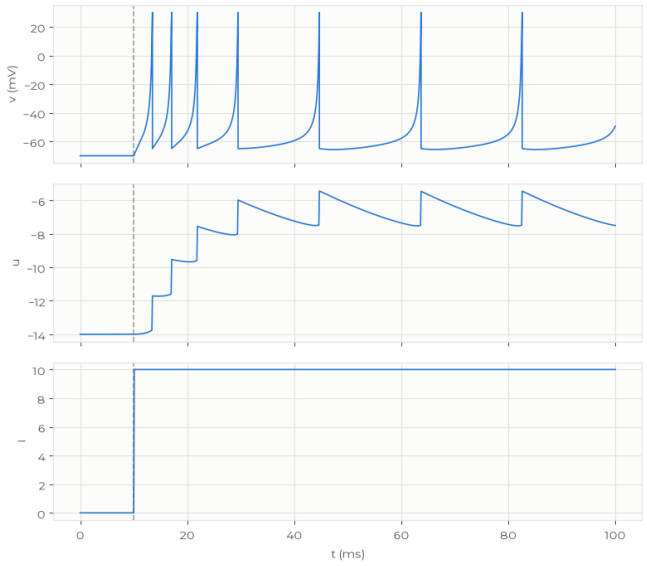


Figure 2. Time evolution of the membrane potential $v(t)$, the recovery variable $u(t)$ and the input current $I(t)$ (from top to bottom).

The thalamo-cortical neuron responds in two different ways depending on the stimulus it receives. In TC1, a weak constant current makes the neuron fire one spike at a time, with well-separated spikes. In TC2, the current is negative: while it is applied, the potential drops below rest, and when it is removed, the neuron responds with a burst of spikes before returning to rest.

Finally, in RZ a small current makes the potential oscillate slightly, but without firing. Only when a brief pulse is applied at 180 ms does the neuron start firing repetitively.

C. Effect of the integration step

Figure 4 shows the effect of the integration step on the base case. All four steps produce seven spikes, but their times are shifted: with the largest step ($h = 0.2$ ms) the spikes are delayed with respect to the reference. This difference grows with the number of spikes, with a marked increase between the fourth and the fifth spike. At the last spike, the shift is 2.5 ms with $h = 0.2$ ms, 0.7 ms with $h = 0.1$ ms and 0.3 ms with $h = 0.05$ ms.

IV. DISCUSSION AND CONCLUSIONS

The simulations qualitatively reproduce the eight patterns of Figure 2 of Izhikevich [4], and their differences can be explained by how the parameters regulate the brake that u exerts on v. A large d strengthens the brake with each spike and produces adaptation (RS), a large a releases it quickly and prevents adaptation (FS), and a high c leaves v close to firing after the reset and generates bursts that repeat or not depending on d (CH and IB). In TC2, the negative current also reduces u, and when it is removed the neuron is left with little brake and fires. RZ shows the bistability between rest and repetitive firing described in the paper. Since the protocols of TC1, TC2

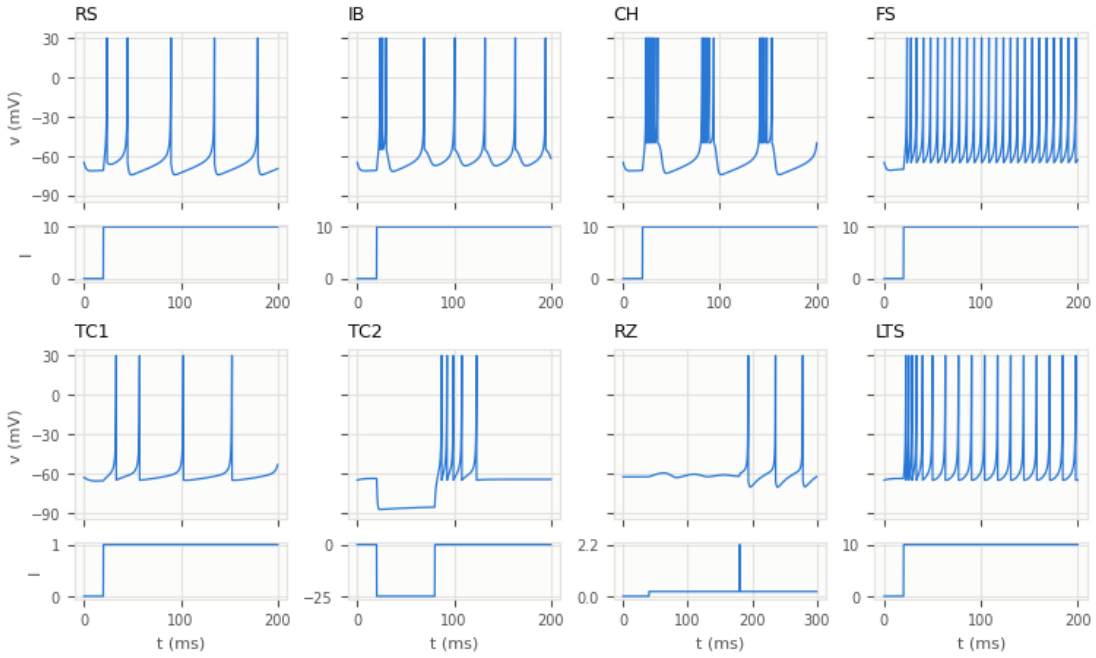


Figure 3. Membrane potential $v(t)$ and input current $I(t)$ (below each panel) for the eight patterns: RS, IB, CH, FS, TC1, TC2, RZ and LTS (parameters and currents in Table I).

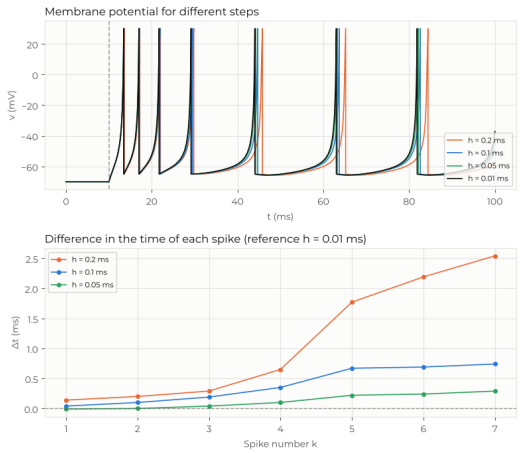


Figure 4. Effect of the integration step on the base case. Top, membrane potential for $h = 0.2$, 0.1, 0.05 and 0.01 ms. Bottom, shift of each spike with respect to $h = 0.01$ ms.

and RZ are specific to this assignment, the comparison in those cases is qualitative, and they show that the same neuron responds differently depending on the stimulus it receives.

In the numerical integration (Figure 4), the main source of error is the reset. The spike is detected at the end of the step in which v exceeds 30 mV, with an error of up to one step h, and since each spike starts from the previous reset, the shifts accumulate. Thus, although RK4 is a fourth-order method, the reset limits the accuracy of the spike times. With $h = 0.1$ ms, the step used in the paper, the spike times depend on the step, but the patterns are not altered.

In summary, with two differential equations and four parameters, the Izhikevich model reproduces a wide variety of neuronal behaviours without describing in detail the mechanisms that give rise to them. It thus lies at the middle point between detail and simplicity put forward by Dayan and Abbott [1], which makes it possible to simulate networks with a large number of neurons [4].

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